

Seminar 10

08.05.2023 10:10

Eigenvalues, Eigenvectors.

1. Find the eigenvalues and eigenvectors of the following matrices in $\text{Mat}_{2 \times 2}(\mathbb{R})$:

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

a) $\dots$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \Rightarrow \lambda_1 = 1, \lambda_2 = -1$$

$$\lambda = -1: \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \forall (x, y) \in \mathbb{R}^2 \mid x=0 = \{0, y\} \mid y \in \mathbb{R} = \langle (0, 1) \rangle \Rightarrow m_g(\lambda) = 1$$

$$\lambda = 1: \begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \forall (x, y) \in \mathbb{R}^2 \mid y=0 = \{x, 0\} \mid x \in \mathbb{R} = \langle (1, 0) \rangle \Rightarrow m_g(\lambda) = 1$$

M is diagonalisable if $\exists S \in GL_n(\mathbb{R})$:

$$M = S^{-1} \Lambda S, \text{ where } \Lambda = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix} \Leftrightarrow \exists \text{ a basis of eigenvectors of } M \text{ for } \mathbb{R}^n$$

$\Leftrightarrow \lambda$ eigenvalue of M

$$m_a(\lambda) = m_g(\lambda)$$

algebraic multiplicity $\rightarrow$ geometric multiplicity

$$b) B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \Rightarrow \lambda_1 = \lambda_2 = 1 \Rightarrow m_a(\lambda) = 2$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \forall (x, y) \mid y=0 = \langle (1, 0) \rangle \Rightarrow m_g(\lambda) = 1 \Rightarrow$$

$\Rightarrow B$ is not diagonalisable

$$c) C = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \Rightarrow \lambda = 1 \text{ with multiplicity } 2$$

Same as above

12. Verify that the matrices

$$A = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{bmatrix} \text{ and } B = \frac{1}{11} \begin{bmatrix} -9 & -2 & 6 \\ 6 & -6 & 7 \\ 2 & 9 & 6 \end{bmatrix}$$

belong to $SO(3)$. Moreover, determine the axis of rotation and the rotation angle.

$A \in M_{n \times n}(\mathbb{R})$ orthogonal if $A^T A = I_n$

$O(n) \rightarrow$ the set of orthogonal matrices

$SO(n) \rightarrow$ the set of special orthogonal matrices

$$A \in SO(n) \Leftrightarrow \begin{cases} A \in O(n) \Leftrightarrow A^T A = I \\ \det A = 1 \end{cases}$$

$$A \in SO(2) \Leftrightarrow A \text{ has the form } \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \theta \in \mathbb{R}$$

$$\textcircled{1} \cos \theta = \frac{\text{tr}(\sin \phi)}{2}, \phi \in \mathbb{R}^2$$

$$\textcircled{2} \cos \theta = \frac{\text{tr}(\sin^2 \phi) - 1}{2}, \phi \in \mathbb{R}^2$$

$$A = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{bmatrix} \Rightarrow \det A = \frac{1}{27} (1 + 4 + 4 + 8 + 8 - 1) = 1 \Rightarrow A \in SO(3)$$

If f is a rotation, then its axis is given by

$$\text{Fix}(f) = \{ (x, y, z) \in \mathbb{R}^3 \mid f(x, y, z) = (x, y, z) \} = \{ (x, y, z) \in \mathbb{R}^3 \mid A \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \}$$

$$\text{tr}(A) = \sum_{i=1}^n a_{ii}$$

$$\text{tr}(A) = \frac{1}{3}(-1 - 2 + 2) = -\frac{1}{3} \Rightarrow \cos \theta = \frac{-\frac{1}{3} - 1}{2} = -\frac{2}{3}$$

$$\theta = \pi - \arccos \frac{2}{3}$$

$$A \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Leftrightarrow A \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$A \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -x + 2y - 2z \\ -2x - 2y - z \\ -2x + y + 2z \end{pmatrix}$$

$$\frac{1}{3} \begin{pmatrix} -x + 2y - 2z \\ -2x - 2y - z \\ -2x + y + 2z \end{pmatrix} - \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \mid \cdot 3$$

$$\begin{pmatrix} -4x + 2y - 2z \\ -2x - 5y - z \\ -2x + y + 2z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -2 & 1 & -1 & 0 \\ -2 & -5 & 1 & 0 \\ -2 & 1 & -1 & 0 \end{pmatrix} \xrightarrow{\substack{L_2 \leftarrow L_2 - L_1 \\ L_3 \leftarrow L_3 - L_1}} \begin{pmatrix} -2 & 1 & -1 & 0 \\ 0 & -6 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \Leftrightarrow \begin{cases} -2x + y - 2 = 0 \\ -6y + 2z = 0 \end{cases} \Leftrightarrow \begin{cases} x = 2 \\ y = 2 \\ z = 3 \end{cases} \Rightarrow \text{Fix}(A) = \langle (-1, 1, 3) \rangle$$

10. Determine the matrix form of a rotation with angle 45° having the same center of rotation as the rotation

$$f(x) = \frac{1}{\sqrt{13}} \begin{bmatrix} 2 & -3 \\ 3 & 2 \end{bmatrix} x + \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

1

$$f(x) = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix} x + \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$\frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \mid \cdot 5 \Rightarrow \text{Find Fix}(f(x))$$

$$\begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 5 \\ -10 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -5 \\ 10 \end{pmatrix}$$

$$\begin{pmatrix} -2 & -4 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -5 \\ 10 \end{pmatrix} \Leftrightarrow \begin{cases} -2x - 4y = -5 \\ 4x - 2y = 10 \end{cases} \Leftrightarrow \begin{cases} x = \frac{5}{2} \\ y = \frac{5}{2} \end{cases}$$

$$R_\theta(P) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$R_\theta \begin{pmatrix} \cos \frac{\pi}{4} & -\sin \frac{\pi}{4} \\ \sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{pmatrix} \cdot \begin{pmatrix} \frac{5}{2} \\ \frac{5}{2} \end{pmatrix}$$

$$R_\theta(P) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \cdot \begin{pmatrix} \frac{5}{2} \\ \frac{5}{2} \end{pmatrix}$$

Rotation with θ of a point P around $A(x_0, y_0)$

$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \cdot [P]_R$$

$$[Q]_R = \sqrt{R_{R'}} \cdot [Q]_{R'} + [A]_R$$

$$[Q]_R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \cdot [P]_{R'}$$

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